

Chromosome-level genome assembly of the common straight swift, *Parnara guttata*, provides insights into adaptive evolution on its rice host

Xuan Zhang,^a Xiangyu Hao,^a Jintian Xiao,^a Hideyuki Chiba^b and Xiangqun Yuan^{a*}



Abstract

BACKGROUND: The common straight swift, *Parnara guttata*, a leaf-feeding rice pest, is widely distributed in Asia, particularly in China, Japan and South Korea. The larvae feed on leaves and bind them together into rolled shelters, reducing photosynthesis and rice heading. However, the lack of a high-quality genome has limited investigation into environmental adaptation in *P. guttata* and the development of integrated pest management strategies.

RESULTS: Here, we generated a chromosome-level reference genome for *P. guttata* by integrating PacBio HiFi long reads with Hi-C technology. The final assembly was 484.88 Mb with a scaffold N50 of 32.96 Mb, and 99.41% of the sequences were anchored to 16 chromosomes. BUSCO analysis showed a completeness of 99.2%. Repetitive sequences accounted for 36.9% of the genome, and 15 241 protein-coding genes were annotated. Comparative genomic analyses indicated expansions in gene families related to muscle development, lipid metabolism, detoxification and immunity. We also found tandem expansions of CYP6/CYP9 and UGT33/UGT39/UGT40 genes, together with positive selection in ATP-binding cassette transporters. Midgut transcriptome analysis further identified several highly expressed candidate detoxification genes, including tandemly duplicated P450 and UGT genes, as well as *PgCCEg206*. By contrast, fewer chemosensory genes were identified in *P. guttata* than the other lepidopteran rice pests.

CONCLUSION: The high-quality *P. guttata* genome provides a valuable resource for studying skipper butterfly evolution and offers new insights into genomic features potentially associated with detoxification and adaptation to its Poaceae host.

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Supporting information may be found in the online version of this article.

Keywords: comparative genomics; host adaptation; midgut transcriptome; *Parnara guttata*; rice skipper

1 INTRODUCTION

The common straight swift, *Parnara guttata*, is a major leaf-feeding rice pest in the family Hesperiidae.¹ In the early instars, larvae typically graze along the leaf margins, which results in characteristic notches. However, from the third instar onward, larvae spin silk to bind single or multiple leaves into a leaf sheath and then feed concealed inside; under heavy infestation, this behavior can impede the emergence of new leaves and panicles and consequently reduce yield (Fig. 1).^{2,3} Food intake is relatively low before the third instar, but it increases obviously after the fifth instar, at which point the larvae may even break off the rice ears.^{3,4}

The number of annual generations of *P. guttata* decreases with increasing latitude. For example, populations in central and southern China can complete approximately five to eight generations per year, whereas there are three generations in northeastern China, Japan and Korea generally.^{2–4} In Iksan (Korea), three

generations occur in rice fields between May and September. It is the second generation of larvae appearing in late July that cause the most severe damage, with reported damage of more than 30% in some cases.² By contrast, in Japan the third generation is typically the most destructive, and outbreaks have frequently been recorded in paddy fields of central and western Japan.^{4,5} *P. guttata* is widely distributed across Asia, including

* Correspondence to: X Yuan, Key Laboratory of Plant Protection Resources and Pest Management of Ministry of Education, College of Plant Protection, Northwest A&F University, Yangling 712100, Shaanxi, China. E-mail: yuanxq@nwsuaf.edu.cn

a Key Laboratory of Plant Protection Resources and Pest Management of Ministry of Education, College of Plant Protection, Northwest A&F University, Yangling, China

b B.P. Bishop Museum, Honolulu, HI, USA



Figure 1. Field symptoms and ecological photographs of *Parnara guttata* on rice plants. (a) Damage symptoms on early-season rice in Xinhua County, Hunan Province, photographed on 27 May 2025. (b) Damage symptoms on late-season rice in Xinhua County, Hunan, photographed on 8 September 2025. (c) Larva feeding on rice leaves. (d) Adult resting on a rice leaf.

Bangladesh, Myanmar, China, India, Nepal, Japan, Korea, Laos, Pakistan, Vietnam and Central Asia, with well-documented migratory behavior.^{5–8} The population dynamics of the *P. guttata* are often characterized by distinct interannual fluctuations and localized outbreaks. In paddy fields of central and western Japan, systematic investigations of stage-specific survival and mortality factors across multiple sites have shown that population densities vary substantially among years and fields. Historical records also suggest that outbreaks in Japan's history show a certain degree of periodicity; for example, records from 1887 to 1953 indicate outbreaks approximately every 6 years.⁹ In Chinese records, *P. guttata* is likewise commonly described as a pest that occurs intermittently and may erupt locally, particularly in mountainous rice-growing regions (Fig. 1).³

Host adaptation in herbivorous insects is shaped by the long-term evolutionary interaction between plant chemical defenses and insect counter-defense systems.^{10,11} Plants produce diverse specialized metabolites that directly or indirectly reduce herbivore performance, while phytophagous insects have evolved behavioral, sensory, digestive and detoxification mechanisms to recognize suitable hosts and overcome host-derived allelochemicals.^{12,13} At the molecular level, this adaptation often involves evolutionary changes in detoxification and chemosensory gene families, including cytochrome P450 monooxygenases (P450s),

UDP-glycosyltransferases (UGTs), glutathione S-transferases (GSTs), carboxyl/cholinesterases (CCEs), ATP-binding cassette transporters (ABCs), odorant receptors (ORs), ionotropic receptors (IRs), gustatory receptors (GRs), odorant-binding proteins (OBPs) and chemosensory proteins (CSPs).^{11,14} These gene families mediate host recognition, xenobiotic metabolism, conjugation and metabolite transport, thereby providing a mechanistic bridge between host-plant use and genome evolution. For example, genome-wide comparisons between the highly polyphagous *Spodoptera litura* and the more specialized model species *Bombyx mori* revealed expansions of P450, CCE, GR and GST gene families in *S. litura*, suggesting that expansion of detoxification and sensory genes may contribute to broad host use and xenobiotic tolerance.¹⁵ Several studies have specifically examined how lepidopteran rice pests adapt to rice or related Poaceae hosts. In *Chilo suppressalis*, comparative transcriptome analysis between rice-associated and water-oat-associated populations showed that host-associated differentiation was linked to transcriptional changes in genes related to detoxification and digestion, suggesting that feeding adaptation involves coordinated physiological and metabolic responses.¹⁶ In the rice leaf folder, *Cnaphalocrocis medinalis*, host switching between rice and wheat altered larval susceptibility to abamectin and chlorpyrifos and changed the activities of detoxification enzymes, including carboxylesterases,

multifunctional oxidases and GSTs, suggesting that host plants can shape detoxification capacity and pesticide response.¹⁷ In *Spodoptera frugiperda*, functional experiments have confirmed that knocking down *SfUGT33F32* and *SfCYP302A1* impaired larval survival and development on rice, providing functional evidence that detoxification genes can contribute to adaptation to rice and other Poaceae hosts.^{18,19} Moreover, comparative analyses between the corn and rice strains of *S. frugiperda* revealed strain-specific differences in detoxification gene families, including CYP, CCE, UGT, GST and ABC genes, suggesting that divergence in detoxification repertoires may be associated with adaptation to distinct host-plant metabolites.²⁰ Together, these studies indicate that adaptation to rice or Poaceae hosts in lepidopteran pests may involve detoxification, chemosensory, digestion and regulation of plant defense responses.

Compared with several moth rice pests, the omics basis of host adaptation in the rice skipper *P. guttata* remains poorly understood. In a previous study, we identified 47 ABC transporter genes in *P. guttata* and found that several ABCC and ABCG members showed strong predicted binding affinities with Cry1Ac and chemical insecticides, whereas ABCD, ABCE and ABCF genes were highly expressed in the larval midgut.²¹ These findings provided preliminary evidence that detoxification-related transporters may be involved in xenobiotic handling in this species. Nevertheless, the previous study focused only on the ABC transporter family. A chromosome-level genome assembly and genome-wide analyses of detoxification, chemosensory and adaptive gene evolution are still required to understand the broader genomic architecture underlying specialization in *P. guttata*.

Here, we generated a chromosome-level reference genome for *P. guttata* by integrating PacBio HiFi long reads sequencing with Hi-C chromatin conformation capture. Furthermore, comparative genomics uncovered signatures of adaptive evolution, and our manual annotation of gene families shed light on the specific expansion patterns of detoxification gene families. Finally, midgut transcriptome analysis further indicated that the detoxification system is likely driven by specific highly expressed members. Several tandemly duplicated P450 and UGT members, broadly expressed GST genes, and the extremely high expression of *PgCCEg206* provided expression-based support for prioritizing candidate genes potentially involved in host-associated detoxification. These results not only serve as a foundational resource for future evolutionary, ecological and functional genomic studies, but also provide candidate clues to genomic features potentially associated with its adaptation to Poaceae host plants.

2 MATERIALS AND METHODS

2.1 Sampling

The samples of *P. guttata* used in this study were collected from rice fields in Chang'an county, Shaanxi, China in September 2022. Pupae were collected from rice fields and subsequently brought back to the laboratory, where they were kept until emergence. Within 6 h after adult emergence, fully wing-expanded adults were collected, and their gender was determined before they were rapidly frozen with liquid nitrogen.

2.2 Genome sequencing

Genomic DNA of *P. guttata* was extracted from one female adult using the Blood & Cell Culture DNA Kit (Qiagen, Hilden, Germany; catalog no. 13323). High-quality genomic DNA was used to construct three libraries. Briefly, these included: (i) a short

reads library for estimating genome size; (ii) a PacBio CCS 15 kb library; and (iii) a Hi-C library. For genome size estimation and Hi-C sequencing, libraries were constructed with an insert size of 350 bp paired-end reads, and sequencing was performed on the Illumina NovaSeq 6000 platform. Meanwhile, both transcriptome and Oxford Nanopore Technologies (ONT) full-length transcriptome data were obtained from the whole body of another female adult. All library construction and sequencing were performed by Berry Genomics (Beijing, China). Finally, a total of 274.06 Gb data was obtained, comprising 108.65 GB of Illumina reads, 102.21 Gb of Hi-C reads, 30.31 Gb of CCS reads, 20.66 Gb of transcriptome reads and 12.23 Gb ONT reads.

2.3 Genome assembly

For Illumina raw data, we used fastp²² with default parameters to perform quality control, removing adapter sequences and low-quality sequences. Genome size and heterozygosity were estimated by GenomeScope v2.0²³ with parameter '-k 19'. Subsequently, the primary assembly of PacBio CCS reads was generated using Hifiasm v0.16²⁴ under default configurations. Three rounds of polishing were conducted by Racon²⁵ for the draft assembly. In the first round, we aligned long reads to the initial assembly using minimap2,²⁶ and in the second and third rounds, BWA v0.7.17-r1188 was used to align short reads to the polished assembly.²⁷ Redundant contigs were removed using Purge_Dups v1.2.5.²⁸ The quality-controlled Hi-C reads were mapped onto the assembly using Juicer v1.6.215.²⁹ Afterwards, contigs were anchored onto chromosomes using 3D-DNA v18092216.³⁰ To enhance accuracy, we manually corrected scaffolding errors, including misjoins, translocations and inversions in Juicebox v1.11.0816.³¹ We used the run-asm-pipeline-post-review.sh script to produce the final assembly. Assembly completeness was evaluated by aligning the final assembly against the insecta_odb10 ($n = 1367$) and lepidoptera_odb10 ($n = 5286$) in BUSCO v5.2.2 software.³²

2.4 Genome annotation

To comprehensively analyze the repetitive sequence landscape of the *P. guttata* genome, we constructed three custom databases. First, we predicted long terminal repeat (LTR) retrotransposons using both LTRharvest³³ and LTR_FINDER,³⁴ followed by filtering the results through LTR_retriever.³⁵ These filtered predictions were then combined with the Miniature Inverted-repeat Transposable Elements (MITEs) predicted by MITE-Hunter³⁶ to create the LTR-MITE.lib library. In the second step, we performed a hard masking of the LTR-MITE.lib using RepeatMasker v4.1.243,³⁷ followed by *de novo* predictions with RepeatModeler v2.0.1.³⁸ To avoid redundancy, the '-LTRStruct' parameter was excluded from this step. The unknown transposable elements (TEs) were compared against the transposase protein database TasesPROT to annotate any unknown TEs. Known and newly annotated TEs were then merged to form the ModelerAll.lib library. In the third step, we extracted homologous sequences of Lepidoptera repeat families from the Dfam5³⁹ and RepBase-20181026⁴⁰ databases (<http://www.girinst.org>) to perform homology-based predictions, resulting in the Homology.db library. We then combined the repeat libraries from different software, LTR-MITE.lib, ModelerAll.lib and Homology.db into allRepeats.lib, removing duplicates before reclassifying unclassified LTRs using DeepTE⁴¹ with the 'Metazoans_model' under the species parameter (-sp M). Finally, RepeatMasker was used with the 'rmbblast' search engine in slow search mode (-s) to annotate the repetitive sequences. In

addition, the Perl scripts calcDivergenceFromAlign.pl and createRepeatLandscape.pl were used to calculate Kimura divergence values and generate the repeat landscape across the genome. Following the annotation of TEs, the hard-masked genome file was split and input into RepeatProteinMask to annotate homologous protein repeat sequences. Tandem repeats were detected via Tandem Repeats Finder software following the parameters 2 7 7 80 10 502 000 -h -d.⁴²

Noncoding RNAs (ncRNA), such as micro RNAs and small nuclear RNAs (snRNA) were identified by cmscan with the Rfam database.⁴³ In addition, transfer RNAs (tRNA) and ribosomal RNAs (rRNA) were predicted by tRNAscan-SE v.2.0.944⁴⁴ and RNAmmer⁴⁵ with default parameters, respectively. These results were merged, redundancy removed, and finally converted to gff format.

For protein-coding genes, *ab initio*, transcript-based, and homology-based analysis were performed on the masked assembly. First, the *ab initio* approach was carried out using PASA (Program to Assemble Spliced Alignment),⁴⁶ Augustus,⁴⁷ SNAP,⁴⁸ GeneID,⁴⁹ and GlimmerHMM.⁵⁰ In short, the assembled transcripts were aligned to the genome assembly, and the gene structure models used for training were refined with PASA. Subsequently, Augustus, SNAP, GeneID and GlimmerHMM were used to predict genes based on these improved training models. Second, RNA sequencing (RNA-seq) reads were first mapped to the reference genome using HISAT2 v2.2.191,⁵¹ followed by transcript assembly with genome-guided assembler StringTie v2.2.19.⁵² The Augustus and geneMarkET⁵³ programs were called for gene annotation by executing the braker.py script. For homology-based gene prediction, protein sequences from nine well-annotated insect genomes were downloaded and mapped to the *P. guttata* assembly. These included seven species obtained from GenBank database: *Danaus plexippus* (GCA_018135715.1),⁵⁴ *Heliconius melpomene* (GCA_000313835.2),⁵⁵ *Papilio polytes* (GCA_000836215.1),⁵⁶ *Papilio xuthus* (GCA_000836235.2),⁵⁶ *Pieris rapae* (GCA_905147735.1),⁵⁷ *Bombyx mori* (GCA_030269925.2),⁵⁸ and *Drosophila melanogaster* (GCA_000001215.4).⁵⁹ In addition, protein evidence for two skipper butterflies, *Lerema accius*⁶⁰ and *Achalarus lyciades*,⁶¹ was acquired from the LepDB database (<http://prodata.swmed.edu/LepDB/>). The protein sequences of related species and their genome annotation file were used as input for the GeMoMa pipeline v1.9^{62,63} with the parameters 'o = true'. GeMoMa_gff_to_gff3.pl was used to convert the output results. All gene evidence identified by the above three approaches was integrated by the EvidenceModeler.⁴⁶ The final gene data set was updated by PASA. Gene function information was obtained from homology searches in public gene databases including NCBI Refseq (NR),⁶⁴ UniprotKB,⁶⁵ eggNOG,⁶⁶ Gene Ontology (GO),⁶⁷ Kyoto Encyclopedia of Genes and Genome (KEGG),⁶⁸ InterPro,⁶⁹ and Pfam.⁷⁰ Finally, to validate the completeness of the chromosome-level assembly scaffolding, the Telomere Identification ToolKit (tidk)⁷¹ was used to identify telomere regions.

2.5 Phylogenetic reconstruction and estimation of divergence time

P. guttata is one of the few butterfly species to have evolved a close association with rice. To investigate the genomic basis underlying its adaptive traits, we selected 13 representative butterfly species spanning the major lineages of Papilionoidea [*P. polytes*, *P. xuthus*, *D. plexippus*, *Kallima inachus* (InsectBase2_ID: IBG_00473),⁷² *Melitaea cinxia* (GCA_905220555.1), *Celastrina argiolus* (GCA_905147145.1),⁷³ *Plebejus argus* (GCA_905404165.1),⁷⁴ *Pieris rapae*, *Colias croceus* (GCA_905220415.1),⁷⁵ *Erynnis tages* (GCA_905147235.1),⁷⁶ *Ochlodes sylvanus* (GCA_905404205.1),⁷⁷

Hesperia comma (GCA_905404135.1)⁷⁸], together with *B. mori* as an outgroup, to establish a comparative evolutionary framework. Because deep phylogenetic divergence among lepidopteran lineages may introduce substantial background genomic variation, comparative analyses were restricted to butterflies to reduce lineage-specific effects and facilitate interpretation of evolutionary changes associated with *P. guttata*. This phylogenetic framework was used to identify lineage-specific genomic features of *P. guttata*, including gene family expansion/contraction and positively selected genes. More targeted comparisons of detoxification and chemosensory gene families were subsequently performed using other lepidopteran rice pests to further evaluate genomic features potentially associated with host adaptation. For each species, the longest transcript of each gene was retained for orthogroup inference using OrthoFinder v2.5.4.⁷⁹ The coding sequences (cds) and peptide sequences (pep) of all single-copy genes were extracted, and then multi-sequence alignment was performed using ParaAT⁸⁰ software with the parameter '-m muscle'. The aligned amino acid sequences were concatenated using SeqKit,⁸¹ and poorly conserved sites were removed using trimAl v1.4,⁸² producing a super-matrix with 1 615 239 sites. Subsequently, the maximum likelihood (ML) phylogenetic tree was inferred by IQ-TREE2.⁸³ It was executed with '-m MFP' to determine the best-fit nucleotide substitution models, '-B 1000' to conduct 1000 ultrafast bootstrap algorithm, and '-bnni' to improve the ultrafast bootstrap tree.

Evolutionary divergence times among the butterfly lineages were inferred using MCMCTree, a tool integrated in the PAML v4.9j software package.⁸⁴ Three secondary calibration points were collected according to the most comprehensive study on the systematics of butterflies: (i) the origin of the entire butterfly lineage was 101.4 Ma (100.0–102.5 Ma, 95% highest posterior density (HPD)); (ii) the crown age of Hesperidae was 73.3 Ma (68.2–75.2 Ma, 95% highest posterior density); and (iii) the crown age of Nymphalidae was 78.8 Ma (77.4–79.8 Ma, 95% highest posterior density).⁸⁵

2.6 Gene family analysis and positive selection analysis

Gene family expansion and contraction were analyzed across the 15 species using CAFE 5⁸⁶ with a *P*-value threshold of 0.05. The time-scaled phylogenetic tree and gene family counts file were used as input files. We set '-p' to specify the root frequency distribution as a Poisson distribution, and set '-k 3' to use the GAMMA model, and use three gamma rates, indicating that different gene families have different rates of evolution. The enrichment analysis of gene families was performed using the omicshare tools (<http://www.omicshare.com/tools>).

Signals of positive selection were detected using the branch-site models implemented in codeml from the PAML package.⁸⁴ The lineage *P. guttata* was designated as the foreground branch. Likelihood ratio tests were performed by comparing the alternative model with the corresponding null model. Resulting *P*-values were corrected for multiple testing using the false discovery rate method, and genes with *q* < 0.05 were considered to be under positive selection.

2.7 Manual annotation of the chemosensory and detoxification gene families

To assess the gene families related to chemical perception and detoxification in *P. guttata*, we manually annotated eight gene families, including CSP, OBP, IR, OR, P450, UGT, GST and CCE. BITACORA pipeline⁸⁷ was used to initially identify these genes as previously described.²³ In brief, this pipeline utilizes the known

high-quality reference protein sequences and Hidden Markov Models to conduct iterative searches in the target genome, correct existing annotations and predict new genes. Gene annotation and classification were further validated using InterProScan and Pfam domain searches. Phylogenetic analyses were then conducted using the confirmed sequences together with those from several major lepidopteran rice pests, *C. suppressalis*, *C. medinalis* and *S. frugiperda*.

2.8 Expression analysis of detoxification-related genes in the larval midgut

To assess the expression patterns of detoxification-related genes in the larval midgut, RNA-seq data from three midgut biological replicates generated in our previous study were re-analyzed.²³ Clean paired-end reads were aligned to the chromosome-level genome assembly of *P. guttata* using HISAT2 v2.2.1.⁵¹ The alignment files were sorted and indexed and gene expression levels were then quantified using StringTie v2.2.1 with the reference genome annotation.⁵² Because several detoxification genes were manually curated during gene family annotation, the curated gene models were merged into the original GFF3 annotation file before expression quantification. TPM values were extracted for four detoxification-related gene families, including P450s, GSTs, UGTs and CCEs, according to the manually curated gene family annotation. For genes with standardized names, original gene IDs were replaced by the corresponding curated gene names. TPM values from the three midgut biological replicates were merged into family-specific expression matrices and transformed using $\log_{10}(\text{TPM} + 1)$ for visualization.

3 RESULTS

3.1 Genome sequencing and assembly

To estimate the genome size of *P. guttata*, 50.5 Gb of Illumina raw reads was used for *k*-mer analysis. The genome size was calculated to be 440.20–445.41 Mb (*k* = 17, 19, 21) (Supporting Information, Figs S1–S3). The *P. guttata* assembly was constructed using 30.3 Gb of HiFi clean reads. The resulting primary assembly was 484.88 Mb, comprising 35 scaffolds with a scaffold N50 of 32.96 Mb after polishing and removal of redundant sequences. Subsequently, 50.1 Gb Hi-C data was used to anchor contigs to scaffolds. We successfully anchored 482.02 Mb of contigs across 16 chromosomes (Fig. 2(a), (b)), consistent with the previously reported karyotype (*n* = 16),⁸⁸ achieving a high assembly completeness of 99.41%. One chromosome exhibited approximately 50% sequencing depth compared with the others, leading us to identify it as the Z chromosome, which was also confirmed by synteny analysis with the other skippers. The presumptive Z chromosome exhibited an average sequencing depth of 37× coverage, significantly lower than the 65× coverage observed for the autosomes.

The quality of our genome assembly was demonstrated through two different methods. First, the mapping rates of PacBio and Illumina reads were 99.48% and 98.41%, respectively. Second, we used BUSCO software to evaluate the completeness of the assembly. Results showed that 99.2% and 98.5% orthologous genes were captured in insecta_odb10 (*n* = 1367) and lepidoptera_odb10 (*n* = 5286), respectively. These results show that the assembly meets high-quality standards (Table 1).

3.2 Gene annotation and repeat analysis

Approximately 36.79% of the *P. guttata* genome was identified as repeat elements, with a total length of 178.39 Mb. The top five repeat classes were DNA transposons (14.95%), LTRs (11.57%),

short and long interspersed nuclear elements (SINE and LINE: 3.66% and 3.44%, respectively), and tandem repeats (0.80%) (Table 2). In addition, we also annotated ncRNA. The results showed that tRNAs had the largest number of 7772 and the longest size (total of 559.26 kb) of all RNA types. The 22S rRNA sequences were divided into three types: 5.8S (133), 18S (48) and 28S (44), and they represented genome lengths of 15 383, 92 772 and 256 436 bp, respectively. Notably, 28S had the longest mean length of all ncRNA classes at 5828.09 bp. For snRNA, there are three major types: CD-box, HACA-box and splicing (Supporting Information, Table S1).

A total of 15 241 protein-coding genes were annotated by the combination of *ab initio*, homolog-based and transcriptome-based annotation. The mean sizes of genes, exons and introns were calculated as follows: 15 045.82, 265.60 and 2256.97 bp (Table 1). Each gene contained an average of 6.86 exons and 5.86 introns. BUSCO evaluation revealed that 96.3% of the 5286 benchmark genes were complete (Table 1). Gene function annotation showed that 11 745 genes (77.06%) and 8726 (57.49%) genes were annotated by InterProScan and the UniProtKB/Swiss-Prot database, respectively. In addition, evolutionary genealogy of genes: Non-supervised Orthologous Groups (eggNOG), Gene Ontology (GO), Clusters of Orthologous Groups/Karyotic Orthologous Groups (COG/KOG) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways were annotated to 10 965, 7935, 10 965 and 4220 genes, respectively.

We identified 29 telomeric regions across 16 chromosomes. Among these, Chr04, Chr13 and Chr14 exhibited telomeric sequences at only one end, with no sequences detected at the opposite end. The remaining chromosomes showed the presence of telomeric regions with 'AACCT' repeats at both ends (Supporting Information, Fig. S4).

3.3 Orthologs and phylogenetic analysis

Protein-coding genes from 14 butterfly species and *B. mori* were preprocessed before orthology inference. First, we extracted the longest transcript per gene, filtered out sequences shorter than 150 bp or 50 amino acids in length, and used the processed sequences as input for OrthoFinder v2.5.4⁷⁹ to infer orthologous groups and species-specific gene families. A total of 293 150 genes were clustered into 23 554 orthologous groups with a mean orthogroup size of 12.4 (Supporting Information, Table S2). Of 5973 orthogroups presented in all species, 3171 single-copy orthologs were identified. The Venn diagram of shared orthologous groups showed that 429 genes in 85 gene families are specific to *P. guttata* (Fig. 2(c)). These genes were mainly related to 'Terpenoid backbone biosynthesis', 'Metabolism of xenobiotics by cytochrome P450', 'Insect hormone biosynthesis' and other pathways when compared against the KEGG database (Fig. 3(a), Supporting Information, Tables S3 and S4).

A highly reliable phylogenetic tree of 14 butterflies and 1 out-group was constructed based on 3171 single-copy orthologous genes (bootstrap values of all nodes for the ML method = 100) (Fig. 2(d)). ML analysis showed that the relationships among butterfly families were consistent with previous studies.⁸⁵ The phylogenetic relationship within Hesperidae was *E. tages* + (*P. guttata* + (*L. accius* + (*O. sylvanus* + *H. comma*))). Molecular dating analyses revealed that Hesperidae split at 98.54 Ma (95% HPD: 93.16–102.44 Ma). Diversification within the family occurred around 62.75 Ma (95% HPD: 54.62–75.2 Ma), marked by the separation of *E. tages* from the other skippers. Finally, *P. guttata* separated from the remaining lineage at 39.85 Ma (26.6–

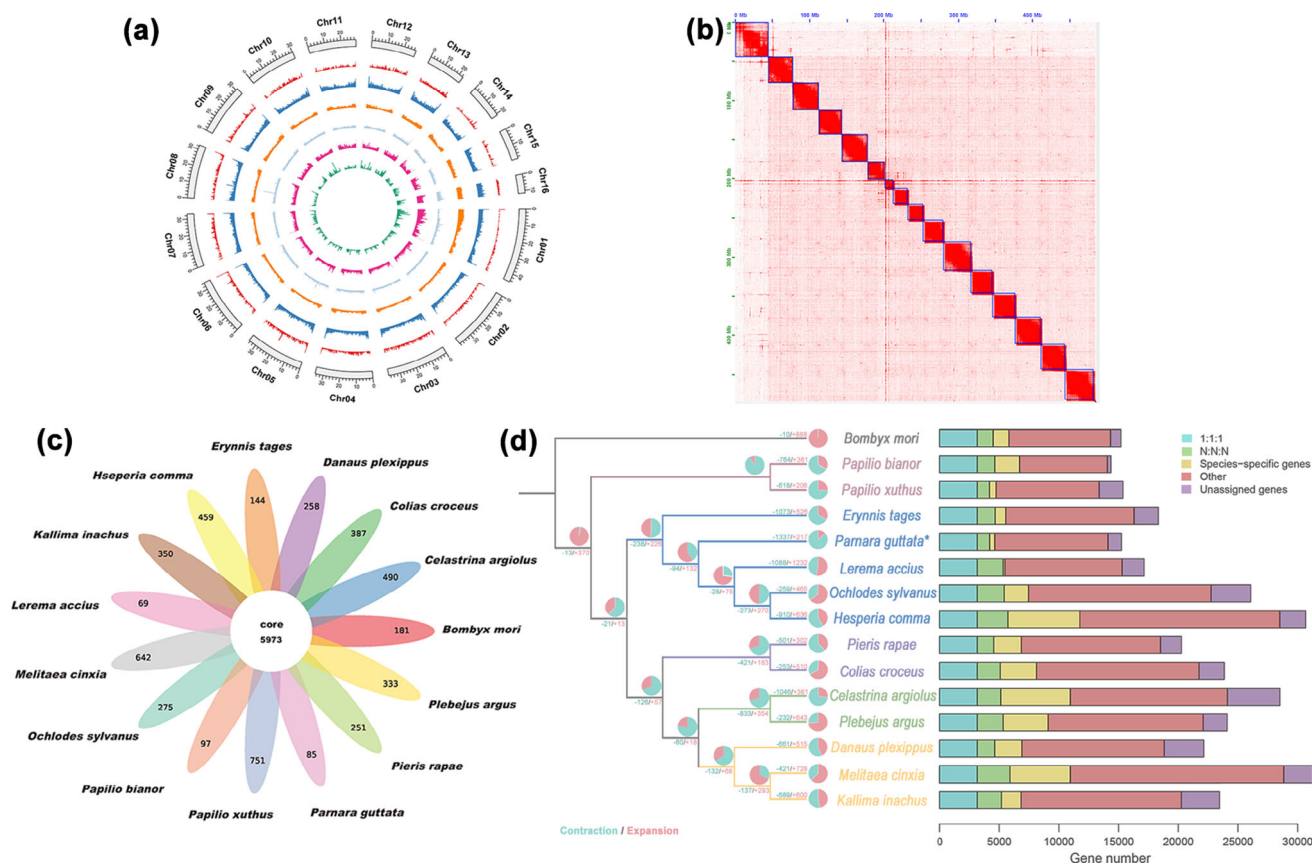


Figure 2. Chromosome-level genome assembly of *Parnara guttata* and phylogenetic relationships of butterflies. (a) Circular plot of the genome showing chromosome length, gene density, GC levels, transposable element distribution, tandem repeat regions, LTR_Gypsy and LTR_Copia content from outside to inside. (b) Hi-C contact map of *P. guttata* assembly. The blue lines correspond to chromosomes. (c) Common and unique gene families between 15 species. The white overlapping parts represent the number of gene families shared between different species and the non-overlapping parts represent the gene families unique to each species. (d) Expansion and contraction analysis of gene families and gene orthology results. The left-hand side shows the phylogenetic relationships and analysis of gene families among 15 species. The phylogenetic tree of *P. guttata* and other insect species was constructed using the maximum likelihood method with concatenated protein sequences of 3171 single-copy orthologous genes with 1000 bootstrap replicates. Gene family expansions (red) and contractions (light blue) are shown as pie charts with numbers denoting family counts. Orthologous gene data are shown in the bar plot.

53.13 Ma, 95% HPD). These results are similar to previous studies (Supporting Information, Fig. S5).^{1,85}

3.4 Gene family and positive selection analysis

CAFE 5⁸⁶ analysis revealed 217 gene families under expansion and 1337 gene families under contraction in *P. guttata* (Fig. 2(d)). After applying a threshold of $P < 0.05$, 60 gene families were found to experience significant expansion, and 171 gene families showed significant contraction, encompassing 654 and 205 genes, respectively. Functional enrichment analysis of the expansion gene families showed that many genes were significantly enriched in the GO term such as 'GO: 0007526 larval somatic muscle development', 'GO: 0007525 somatic muscle development' and 'GO: 0002164 larval development', which may be associated with the strong body and muscular physique of *P. guttata*. GO terms including 'GO: 0016726 oxidoreductase activity, acting on CH or CH2 groups, NAD or NADP as acceptor' and 'GO: 0016623 oxidoreductase activity, acting on the aldehyde or oxo group of donors, oxygen as acceptor' were also enriched, potentially reflecting metabolic activities related to wing movement. Terms such as 'GO:0015020 glucuronosyltransferase activity', 'GO: 0008194 UDP-glycosyltransferase activity', 'GO:0006011 UDP-glucose metabolic process' were enriched and may be involved in

detoxification processes. (Fig. 3(b); Supporting Information, Table S5). Similarly, KEGG functional enrichment also revealed functional categories related to cell cycle regulation, energy metabolism, detoxification metabolism and immune response in *P. guttata*. For example, pathways such as 'Fatty acid biosynthesis and metabolism' may be associated with energy conversion, while 'Drug metabolism-cytochrome P450' and 'Metabolism of xenobiotics by cytochrome P450' suggested the presence of metabolic capacities that could contribute to tolerance of environmental compounds. Enriched pathways including 'FoxO signaling pathway', 'Porphyrin metabolism' and 'AMPK signaling pathway' may be relevant under conditions of high metabolic demand, and 'Retinol metabolism' may be related to environmental sensing (Fig. 3(c); Supporting Information, Table S6).

Positive selection plays a crucial role in the development of new functions during species evolution. In our study, we identified 352 genes in *P. guttata* that exhibited significant positive selection ($q < 0.05$). These genes were primarily enriched in GO terms such as 'GO: 0005694 chromosome', 'GO: 1903827 regulation of cellular protein localization', 'GO: 0071456 cellular response to hypoxia' and 'GO: 0042127 regulation of cell proliferation' (Supporting Information, Table S7). KEGG enrichment analysis further identified pathways including 'Sphingolipid signaling pathway', 'ABC

Table 1. Summary statistics of *Parnara guttata* genome assembly and gene annotation

Features	Value
Genome assembly	
Size (Mb)	484.88
Number of scaffolds	35
Number of chromosomes	16
Scaffold N50 length (Mb)	32.96
GC (%)	35.67
BUSCO completeness (%)	99.2/98.5 ^a
Genome annotation	
Total number of PCG	15 241
Average PCG length (bp)	15 045.82
Total number of cds	103 141
Average cds length (bp)	223.37
Average number of cds per gene	6.77
Total number of exons	104 541
Average exon length (bp)	265.60
Average number of exons per gene	6.86
Number of genes with introns	13 470
Total number of introns	89 300
Average intron length (bp)	2256.97
Average number of introns per gene	5.86
BUSCO Completeness (%)	96.3 (lepidoptera_odb10)

a 99.2% of insecta_odb10 database, 98.5% of lepidoptera_odb10. cds, coding sequences; PCG, protein coding sequences.

Table 2. Summary statistics of repeat elements of *Parnara guttata* genome

Repetitive element	Total length in bp (%)
DNA	72 581 917 (14.95)
LINE	16 671 566 (3.44)
SINE	17 730 824 (3.66)
LTR	56 049 835 (11.57)
TandemRepeat	3 878 084 (0.80)
Unclassified	15 897 589 (3.28)
Total	178 390 968 (36.79)

transporters' and 'Wnt signaling pathway', which were involved in embryonic development and environments response processes (Fig. 3(d); Supporting Information, Table S8). Among the positively selected genes, *PgABCB3*, *PgABCC5* and *PgABCG4* were found to have undergone positive selection. ABCB, ABCC and ABCG subfamilies represent the principal members of the ABC transporter family involved in environmental adaptation. The ABCB and ABCC subfamilies predominantly mediate the transport of xenobiotic degradation products and contribute to the development of resistance against agricultural pesticides and Bt (*Bacillus thuringiensis*) toxins. By contrast, the ABCG subfamily is primarily responsible for the transport of cuticular hydrocarbons (CHCs) and other wax components. The detection of positive selection in *PgABCB3*, *PgABCC5* and *PgABCG4* may indicate functional diversification in these transport systems during the evolutionary history of *P. guttata*.

3.5 Chemosensory and detoxification gene families in *P. guttata*

Given that the chemical perception system is crucial for survival, reproduction and mating, we identified the OR, IR, OBP and CSP gene families. The results showed that the copy numbers of OBP, IR and OR genes in *P. guttata* were lower than those of other lepidopteran rice pests, with only 28, 20 and 44 genes (Table 3). This may reflect differences in the degree of olfactory reliance between butterflies and moths for host localization and mating, as butterflies potentially depend more heavily on visual cues. Furthermore, *S. frugiperda* possessed the highest number of OR and IR genes, which likely correlates with its highly polyphagous feeding habit; unlike *P. guttata*, which is almost exclusively restricted to Poaceae, *S. frugiperda* can infest more than 300 plant species across families such as Asteraceae, Fabaceae and Solanaceae. In addition, the similar number of CSP genes among these lepidopteran rice pests suggested that the CSP family is more evolutionary conserved, maintaining a baseline level of diversity across various lepidopterans to perceive environmental chemical signals (Table 3).

In terms of detoxification genes, *P. guttata* possessed 75 P450 genes. Although this number was lower than that of *S. frugiperda* (110) and the outgroup silkworm (80), it was higher than those of other rice pests *C. medinalis* (66) and *C. suppressalis* (62) (Table 3). Consistent with the results of gene family expansion, the phylogenetic analysis also revealed an expansion of PgP450 genes, particularly in the CYP3 clan. Chromosomal localization analysis showed that these 75 P450 genes were distributed across 15 chromosomes, except Chr14, among which Chr08 harbored the most PgCYP genes (12) (Fig. 4). These expanded genes formed gene clusters through tandem duplication—such as *PgCYP324A84–PgCYP324A89*, *PgCYP9A364–PgCYP9A366*, *PgCYP6AE248–PgCYP6AE251* and *PgCYP6AB385–PgCYP6AB388*. These patterns suggest that tandem duplication may have contributed to the diversification of detoxification-related genes in *P. guttata*.

The genome of *P. guttata* contained 29 UGT genes, a number comparable with that of two other major rice pests, *C. medinalis* (27) and *C. suppressalis* (30) (Table 3). Phylogenetic analysis classified these genes into ten families, comprising the UGT 33, 34, 39, 40, 41, 42, 44, 45, 46 and 50 families (Fig. 5). UGT40 and UGT33 were the most expanded families, containing ten and eight genes, respectively. The UGT39 family contained three genes, and the UGT46 family included two members; the remaining families were highly conserved, each represented by a single gene. Chromosomal localization analysis revealed that genes of the expanded UGT40, UGT33 and UGT39 families were organized into tandemly duplicated gene clusters. All ten UGT40 genes were located on Chr09, tandemly arrayed in a single cluster. The UGT33 genes were exclusively distributed on Chr02, forming two separate clusters comprising five and three genes, respectively. The three UGT39 genes situated one tandem cluster on Chr04 (Fig. 5).

3.6 Expression profile of detoxification gene families in the midgut

To investigate the transcriptional activity of detoxification-related genes in the larval midgut, we examined the expression profiles of four major detoxification gene families, including P450s, GSTs, UGTs and CCEs (Fig. 6). The three biological replicates showed generally consistent expression patterns. P450 genes displayed strong member-specific expression, with several CYP6 and CYP9 genes showing relatively high expression levels (Fig. 6(a)). Notably,

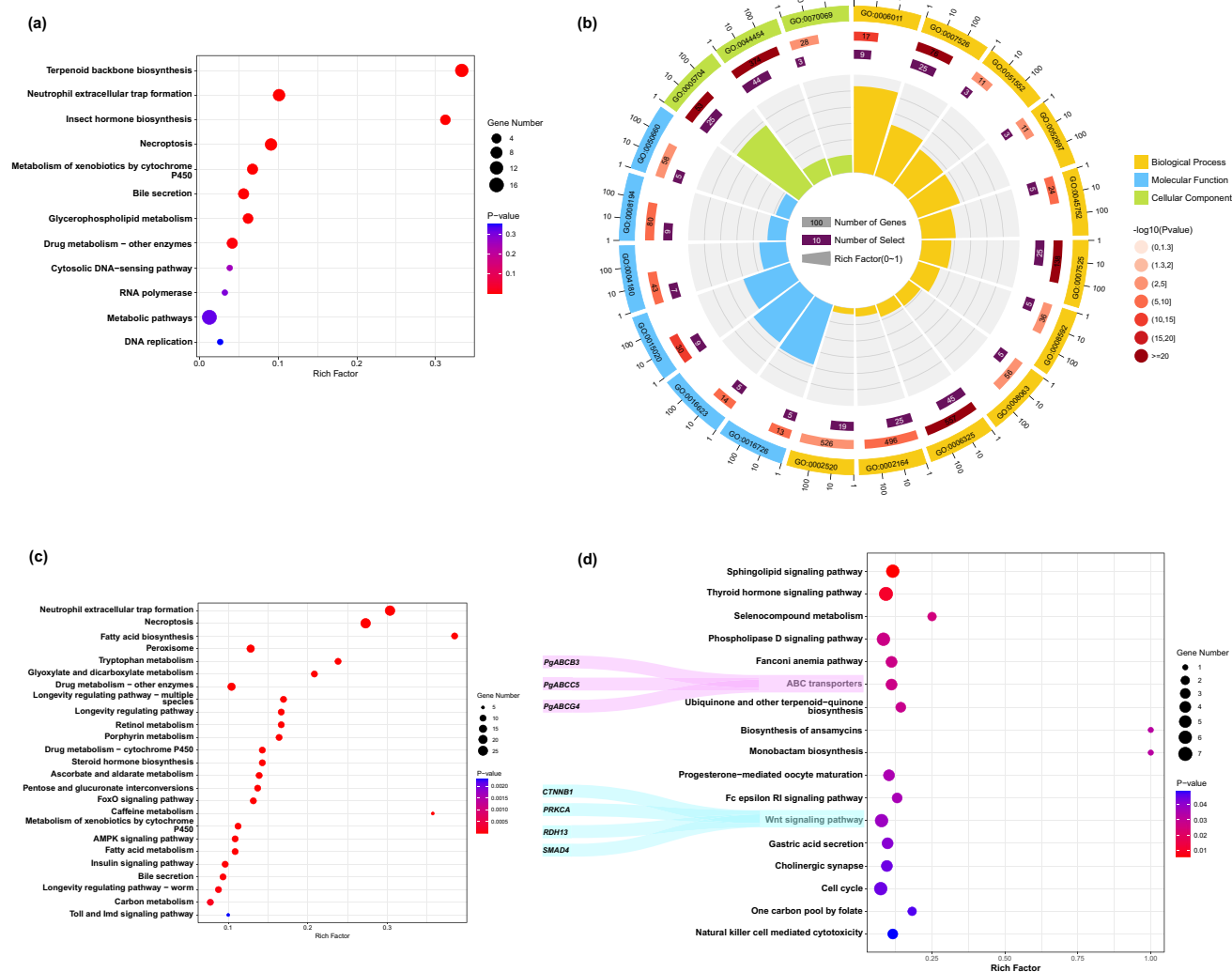


Figure 3. Functional enrichment of species-specific gene families, expansion gene families and positive selection genes in *Parnara guttata*. (a) Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment of species-specific gene families in *P. guttata*. The x-axis represents the Rich Factor, and the y-axis lists the enriched KEGG pathways. Bubble size indicates the number of genes involved in each pathway, and color represents the P-value, with red indicating greater statistical significance. (b) Circular Gene Ontology enrichment plot of significantly expanded gene families. From outer to inner: ring 1, enriched categories (different colors represent distinct categories), with an external coordinate scale indicating gene counts; ring 2, background gene count for each category and the P-value (longer bars indicate more genes; redder colors indicate smaller P-values); ring 3 total number of foreground genes; ring 4, Rich factor for each category (ratio of foreground to background genes), where each background grid interval denotes 0.1. (c) KEGG pathway enrichment bubble plot of significantly expanded gene families. (d) KEGG pathway enrichment bubble plot of positively selected genes. The Sankey diagram on the left links representative positively selected genes (e.g. *PgABC83*, *PgABCC5*, *PgABCG4*, *CTNNB1*, *PRKCA*, *RDH13*, and *SMAD4*) with their associated KEGG pathways, highlighting potential roles of these genes in signaling pathways, detoxification, and environmental adaptation.

Table 3. Chemosensory and detoxification genes number in rice pests and *Bombyx mori*

	OBP	CSP	IR	OR	P450	UGT	CCE	GST
<i>Parnara guttata</i>	28	22	20	44	75	29	67	26
<i>Spodoptera frugiperda</i>	33	20	45	72	110	47	94	28
<i>Cnaphalocrocis medinalis</i>	32	23	19	55	66	27	78	25
<i>Chilo suppressalis</i>	39	21	36	47	62	30	66	21
<i>Bombyx mori</i>	43	16	28	69	80	44	86	23

several highly expressed P450 genes were arranged in tandemly duplicated clusters, including *PgCYP324A84*, *PgCYP324A88*, *PgCYP324A89*, *PgCYP9A364*–*PgCYP9A366*, *PgCYP6AB386* and

PgCYP6AB387, suggesting that tandem duplication may have generated transcriptionally active candidate detoxification genes in the larval midgut. Because CYP6 and CYP9 subfamilies are frequently

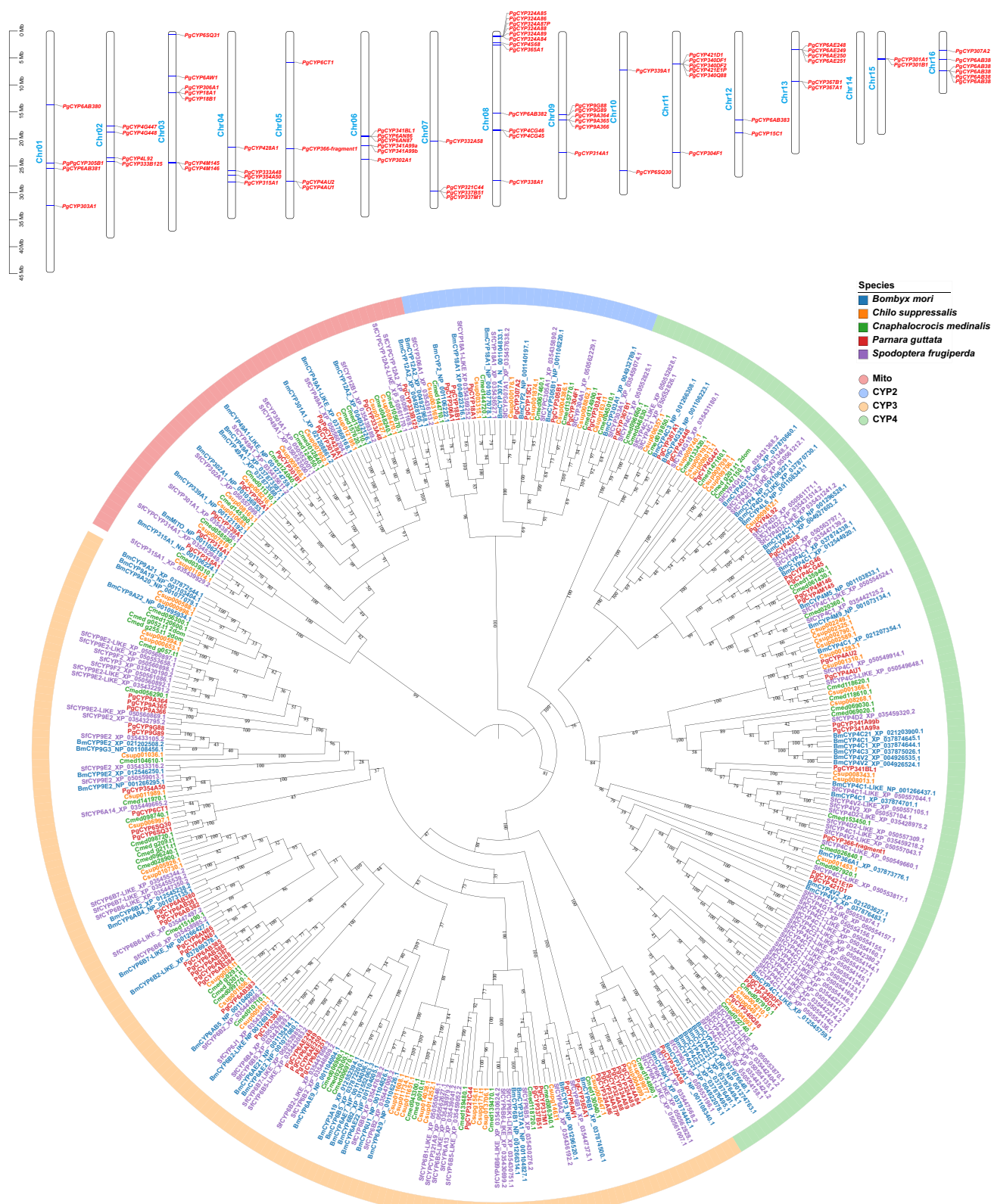


Figure 4. Phylogenetic relationships and chromosomal distribution of cytochrome P450 genes among five lepidopteran species. Upper panel: chromosomal distribution of P450 genes in *Parnara guttata*, with gene positions marked along each chromosome. Clusters of closely located genes indicate tandem duplication events, particularly in the CYP3 clan (e.g. CYP6 and CYP9 subfamilies), suggesting lineage-specific expansion of detoxification-related genes in *P. guttata*. Lower panel: phylogenetic relationship of P450 genes in lepidopteran rice pests and the model insect silkworm.

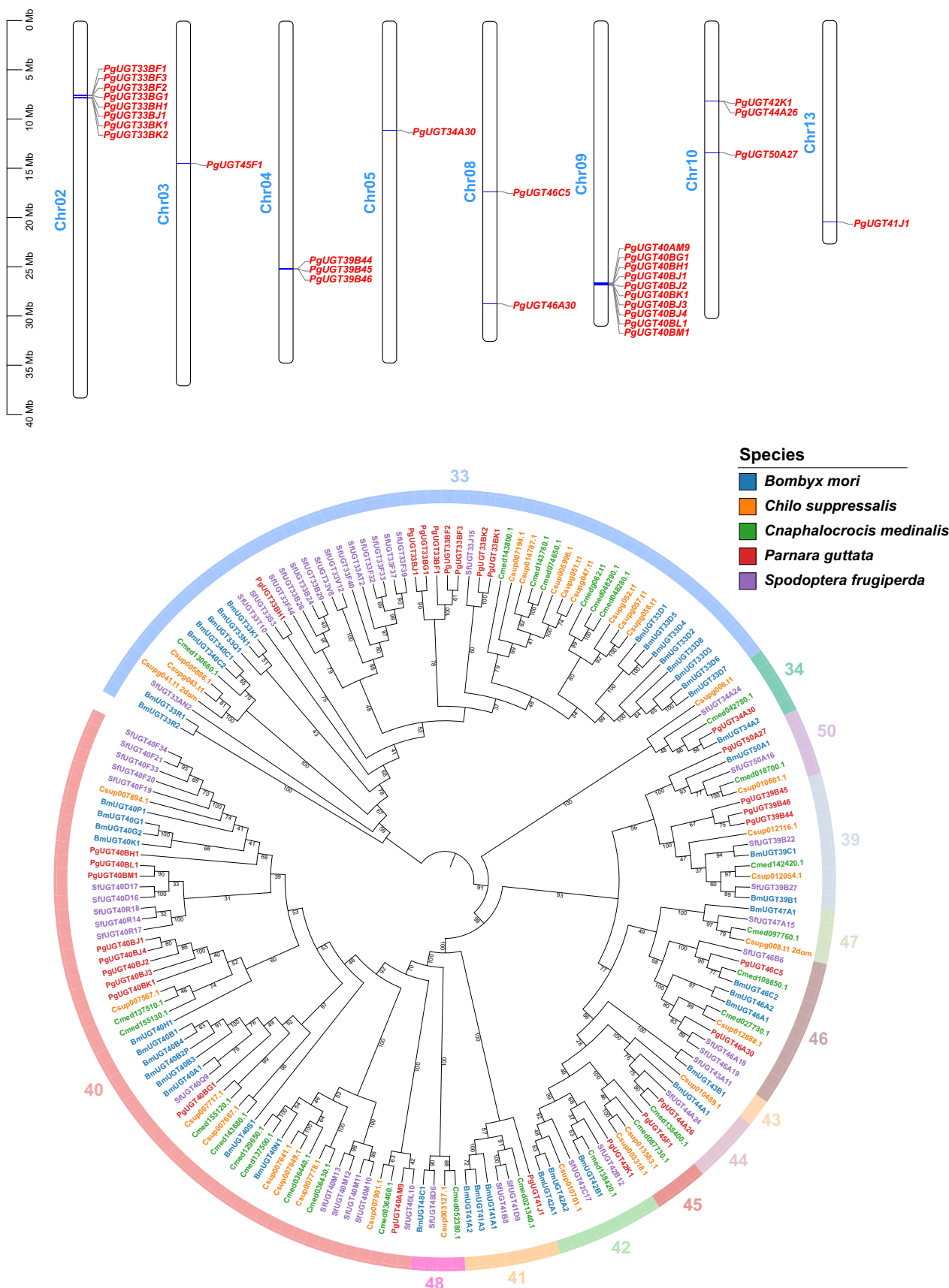


Figure 5. Phylogenetic relationships and chromosomal distribution of UDP-glycosyltransferases among five lepidopteran species. Upper panel: chromosomal distribution of UGT genes in *Parnara guttata*, with gene positions marked along each chromosome. Clusters of closely located genes indicate tandem duplication events. Lower panel: phylogenetic relationship of UGT gene systems in lepidopteran rice pests and the model insect silkworm.

associated with xenobiotic metabolism in lepidopteran insects, these highly expressed tandem duplicates may represent important candidate genes involved in the oxidative metabolism of rice-derived allelochemicals or environmental xenobiotics during feeding.

GST genes displayed a broader expression pattern in the larval midgut (Fig. 6(b)). In contrast to the highly member-specific pattern observed in P450s, many GST genes showed detectable expression across the three biological replicates, suggesting that GST-mediated conjugation and oxidative stress responses may represent a general component of the midgut detoxification system.

By contrast, most UGT genes showed low expression, whereas only a few members, such as *PgUGT33BF2*, *PgUGT33BF3*, *PgUGT40BG1*, *PgUGT40BH1*, *PgUGT40BJ1*, *PgUGT40BJ3* and *PgUGT42K1*, were relatively highly expressed, suggesting member-specific functional specialization in the UGT family (Fig. 6(c)). The restricted expression of UGT genes also implied potential functional specialization among UGT members, with some genes possibly acting in other tissues, developmental stages or under specific induction conditions.

Most CCE genes were detectable in the midgut, and several members showed particularly high expression levels, indicating that CCEs may contribute broadly to hydrolytic detoxification in the larval midgut (Fig. 6(d)). Notably, *PgCCEg206* exhibited extremely high expression compared with other CCE genes and maintained consistently strong expression across all three replicates. This strong and stable expression suggested that *PgCCEg206* may represent a key candidate CCE gene involved in hydrolytic metabolism in the larval midgut. Other highly expressed CCE members may also contribute to the hydrolysis of ester-containing plant compounds or agricultural chemicals.

Together, these results suggested that the detoxification system in the larval midgut of *P. guttata* is likely driven by specific highly expressed members of each detoxification gene family rather than by uniform expression of entire families. The high expression of several tandemly duplicated P450 and UGT genes and the extremely high expression of *PgCCEg206* provided candidate expression evidence for prioritizing genes potentially involved in host-associated detoxification. However, further functional validation will be required to determine their precise roles in the metabolism of rice-derived secondary metabolites or pesticide tolerance.

4 DISCUSSION

High-quality and contiguous genomic resources provide an essential foundation for studies of morphology, damage mechanism, adaptation, chemical communication, migration and resistance evolution of pests. This study presented a high-quality chromosome-level genome of the pest *P. guttata*, which is widely distributed in rice-growing areas across various Asian countries, adding another important component to the genomic resources of lepidopteran rice pests. The genome size of *P. guttata* was 484.88 Mb, which is comparable with the genome sizes of other species in the family Hesperidae.^{76–78} The scaffold N50 of 32.96 Mb was almost the longest among hesperiid genomes studied to date, with the exception of *Gegenes nostradamus*, indicating a high degree of genome continuity. By integrating Hi-C sequencing, 99.41% of sequences were scaffolded onto 16 chromosomes, comprising 15 autosomes and 1 Z chromosome. A total of 29 telomeres were identified, with 3 chromosomes having telomeres at only one end. The completeness of the assembly and annotation was 98.5% and 96.3%, respectively, by BUSCO compared against

the lepidoptera_odb10 ($n = 5286$) data set. This suggests that our genome assembly exhibits high continuity and completeness, which can be attributed to the relatively high sequencing depth achieved in this study. This resource will facilitate future studies of *P. guttata* survival strategies and resistance evolution in agricultural ecosystems. In addition, given *P. guttata*'s distribution across both the Oriental and Palearctic regions, its genome offers a unique opportunity to explore the genetic adaptations and evolutionary dynamics that have enabled this butterfly to thrive in diverse ecological niches.

The detoxification system of xenobiotics in insects is generally regarded as a continuous metabolic system consisting of redox reactions (phase I), conjugation reactions (phase II) and transmembrane transport excretion (phase III).¹⁴ In this study, we found that the key detoxification gene families involved in phase I and phase II detoxification have undergone significant specific expansions, namely P450s and UGTs, respectively. *P. guttata* mainly feeds on Poaceae plants. This feeding habit suggests a long-term ecological association with a Poaceae host, although direct functional evidence for co-adaptation remains to be established. Poaceae plants (such as rice) have various secondary metabolites, including phenolic acids, flavonoids and alkaloids that are defensive chemical substances.⁸⁹ Notably, the expanded P450 genes in this study were mainly concentrated in the CYP6 and CYP9 subfamilies and formed multiple tandemly arranged gene clusters. Formation of such gene clusters may, on the one hand, increase the overall expression level of detoxification enzymes through dosage effect, and on the other hand, expand the substrate recognition range of enzymes through neofunctionalization or subfunctionalization, thereby providing a candidate mechanism for enhancing metabolism of Poaceae-derived defense compounds.

Among lepidopterans, the CYP6 and CYP9 subfamilies are regarded as P450 groups that have undergone the most frequent expansion and possess the highest functional diversity.⁹⁰ For instance, in *S. litura*, expression of CYP9A40 could be induced by coumarins, neem compounds and chlorpyrifos, indicating its dual role in the detoxification of plant chemicals and pesticides.⁹¹ In *Helicoverpa armigera*, CYP6AE not only participated in the metabolism of various insecticides such as imidacloprid, pyrethroids and cypermethrin, but also responded to toxic compounds derived from plants such as gossypol.⁹² In *Cydia pomonella*, *CpCYP9A120/121/122* showed significant functional redundancy in detoxifying the highly effective chlorfenvinphos.⁹³ These studies suggest that expansion of the CYP6/CYP9 subfamilies is usually closely related to the insects' adaptation to plant secondary metabolites and pesticide stress. In our data, several CYP6 and CYP9 genes showed relatively high expression in the larval midgut, including tandemly duplicated genes such as *PgCYP324A84*, *PgCYP324A88*, *PgCYP324A89*, *PgCYP9A364–PgCYP9A366* and *PgCYP6AB386–PgCYP6AB387*. Therefore, the high expression of tandemly duplicated CYP6/CYP9 genes in *P. guttata* may represent a candidate mechanism by which the rice skipper enhances oxidative metabolism of rice-produced secondary metabolites or agricultural xenobiotics.

The UGT gene family is an important component of the phase II detoxification in insects.¹⁴ It mainly converts lipid-soluble compounds into more easily excreted hydrophilic complexes through glycosylation reactions.⁹⁴ Similar to P450, the number of insect UGT genes shows significant dynamic changes among different species and is considered to be closely related to the range of host plants.¹⁸ In Lepidoptera insects, UGT33 and UGT40 are the two

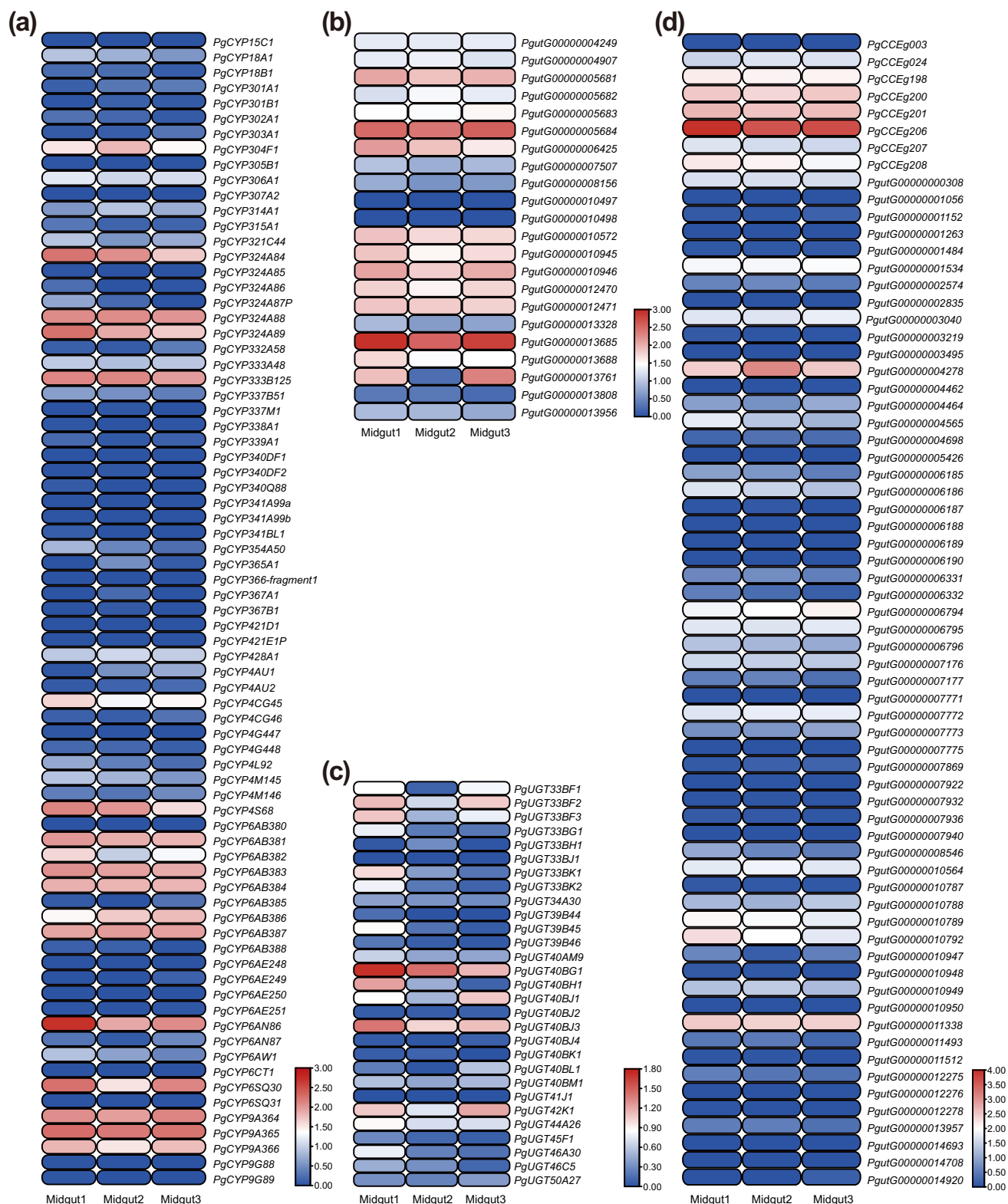


Figure 6. Expression profiles of detoxification-related gene families in the midgut of *Parnara guttata*. Heatmaps show log₁₀-transformed expression values [log₁₀ (TPM + 1)] across three biological replicates (Midgut1–3). (a) Cytochrome P450 (CYP) genes. (b) Glutathione S-transferase (GST) genes. (c) UDP-glucuronosyltransferase (UGT) genes. (d) Carboxyl/cholinesterase (CCE) genes.

subfamilies that frequently undergo tandem duplication. For example, in *S. frugiperda*, knockout of the UGT33 gene cluster significantly reduced the growth ability of larvae on Poaceae crops

(rice, corn and wheat), and significantly increased the sensitivity to the rice-derived alkaloid gramine, indicating that this gene cluster plays a crucial role in adaptation to Poaceae.¹⁸

Subsequently, the five genes (*SfUGT40R14*, *SfUGT40F19*, *SfUGT40M10*, *SfUGT40L10* and *SfUGT40R17*) in the UGT40 gene cluster of *S. frugiperda* were demonstrated to be capable of metabolizing various pesticides, playing a significant role in the evolution of the insect's resistance. In *C. suppressalis*, *CsUGT33AG3* and *CsUGT40AL1* showed significant overexpression in resistant strains, further supporting a potential function of UGT genes in insecticide resistance.⁹⁵ The midgut expression pattern of UGT genes observed in this study was consistent with recent findings in *S. frugiperda*, where only a small number of UGT genes were highly expressed under both artificial diet and maize-feeding conditions, whereas most other UGT genes showed low expression levels.⁹⁶ Thus, the limited number of midgut-expressed UGTs in *P. guttata* may indicate functional specialization, with only particular members contributing to glycosylation-mediated detoxification during rice feeding.

In the third stage of the detoxification system, ABC transporters drive the transmembrane transport of various metabolites through ATP hydrolysis, which is an important mechanism for insects to export toxic metabolites.²¹ ABC transporters are one of the largest transporter families in the insect genome and are involved in various physiological processes such as morphogenesis, pigment deposition, immune regulation, CHC transport and insecticide resistance.^{21,97} ABCB and ABCC mainly participate in detoxification by transporting insecticide metabolites, whereas ABCG may reduce the penetration toxicity of pesticides by enhancing the epidermal barrier.⁹⁷ In addition, ABCB and ABCC have been identified as receptor proteins for *Bt* toxins, and mutations in these proteins can lead to *Bt* resistance.⁹⁸ In this study, significant positive selection signals were observed for *PgABCB3*, *PgABCC5* and *PgABCG4* in *P. guttata*. These selection signals may reflect candidate changes associated with increased metabolite transport demand or long-term exposure to agricultural xenobiotics.

Overall, our results suggested that the detoxification system of *P. guttata* may have undergone coordinated co-evolution across multiple metabolic phases: phase I and phase II detoxification enzymes expanded their metabolic capacity through tandem duplication, while the phase III transport proteins may have contributed to changes in metabolite transport or excretion efficiency. Midgut expression profiles further indicated that this detoxification system is unlikely to be driven by uniform expression of entire gene families, but rather by a subset of highly expressed candidate genes, including P450s involved in oxidation, GSTs and UGTs involved in conjugation, and CCEs involved in hydrolysis. These multilevel molecular adaptations may contribute to the ability of *P. guttata* to tolerate the chemical defenses of Poaceae host plants and the pesticide pressure present in agricultural ecosystems.

5 CONCLUSION

In summary, we generated a high-quality chromosome-level genome assembly for *P. guttata*, providing an important genomic resource for studying the evolution and ecological adaptation of rice pests. Comparative genomic analyses identified candidate adaptive signatures associated with muscle development, energy metabolism and xenobiotic detoxification. Notably, tandem expansions in *CYP6/CYP9* and *UGT33/UGT39/UGT40* genes, together with positive selection acting on several ABC transporters, may represent candidate evidence for coordinated evolution of the detoxification system that could enhance the

metabolic and transport capacity for plant allelochemicals and agricultural pesticides. Midgut transcriptome analysis further indicated that the detoxification system is likely driven by specific highly expressed members. Several tandemly duplicated P450 and UGT members, broadly expressed GST genes, and the extremely high expression of *PgCCEg206* provided expression-based support for prioritizing candidate genes potentially involved in host-associated detoxification. By contrast, the relatively reduced number of chemosensory genes indicates that *P. guttata* may rely less on expanding sensory perception and more on strengthening metabolic detoxification to cope with Poaceae host plants and pesticide pressure in agricultural environments. This study not only provides an important genomic reference for Hesperidae but also offers new insights into the candidate molecular mechanisms underlying host-associated detoxification and adaptive evolution in this rice pest.

The number of annotated skipper genomes remains limited, which currently restricts dense taxonomic sampling in Hesperidae. As additional high-quality skipper genomes and standardized gene annotations become available, future comparative analyses will allow more comprehensive investigations of lineage-specific adaptation, host specialization and the genomic basis of ecological diversification in this family. Given the lack of direct functional assays, these findings should be interpreted as candidate genomic and transcriptomic evidence. Furthermore, future research should combine functional validation with the comparative genomics of Lepidoptera, which will help elucidate how detoxification ability, sensory perception and host-plant chemical components interact to shape the evolutionary strategies of herbivorous insects.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in National Center for Biotechnology Information at <https://www.ncbi.nlm.nih.gov/>, reference number PRJNA1253353.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

XZ and XY conceived the idea and collected the specimens. XZ and XH performed the analysis. JX drew the pictures and collected the data. XZ drafted the manuscript. HX, HC and XY reviewed the manuscript. XY procured funding.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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